Soil Organic Carbon

Sequential Fractionation Protocol

Overview

This protocol isolates four soil organic matter fractions from 10–20 g of air-dried, <2 mm sieved soil. Processing 6–10 samples requires approximately 5 work days.

G.O. Abagandura et al. Geoderma 442 (2024) 116776

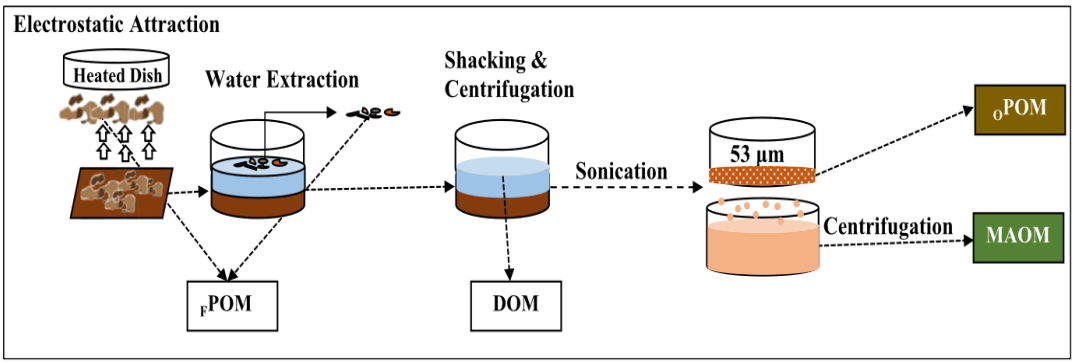

Fig. 2. Procedure for fractionating soil to free particulate organic matter ($_F$POM), dissolved organic matter (DOM), macro-aggregate occluded particulate organic matter ($_O$POM), and mineral-associated organic matter (MAOM), adapted from Jilling et al. (2020). First, $_F$POM was separated by electrostatic attraction and water extraction. Second, the remaining portion of the soil sample and the solution were then shaken, centrifuged and the supernatant was considered as DOM and the pellet remaining was subjected to an ultrasonication and wet sieved using a 53-µm sieve to separate $_O$POM from MAOM.

Part	Fraction	Size Cut	Method	After Isolation
I	Dry Free POM (FPOM)	>53 µm	Electrostatic attraction	Freeze dry → weigh → store
II	Wet Free POM (FPOM)	>53 µm	Pipette wet-sieving	Freeze dry → weigh → store
II	Dissolved OM (DOM)	<0.45 µm	Shaking + centrifuge + 0.45 µm filter	Volume → refrigerate ≤2 wk or freeze-dry
III	Occluded POM (OPOM)	>20 µm	Ultrasonication (60 J/cm³) + sieving	Freeze → freeze dry
III	Mineral-Associated OM (MAOM)	<20 µm	Ultrasonication (60 J/cm³) + sieving	Freeze → freeze dry

⚠ **NOTE:** All water used throughout this protocol is academic-grade DI water.
Low humidity is required for Part I (electrostatic attraction).
Use tinfoil wherever contact surfaces could introduce carbon contamination.

PART I — Dry Free POM (Electrostatic Attraction)

Goal: Remove coarse (>250 µm) free organic particles from the soil using a charged plastic petri dish. The result is (1) Dry FPOM in a 100 mL cup and (2) cleaned bulk soil for Part II.

Materials

- Labeled tin (1/sample)
- 100 mL cup (1/sample)
- Petri dishes
- 250 µm sieve (3" diameter)
- Tinfoil
- DI water squirt bottle
- Hair dryer
- Cotton cloth
- Forceps
- Analytical balance

Steps

1. **Weigh soil.** Tare a labeled tin. Add 20.00 g of air-dried, <2 mm sieved soil.
2. **Sieve.** Sieve soil through 250 µm sieve onto tinfoil. Spread the >250 µm fraction evenly on a separate piece of tinfoil. Do not crush aggregates.
3. **Charge petri dish.** Warm a plastic petri dish with a hair dryer for 10–15 seconds (do not melt). Rub vigorously with cotton for ~10 seconds.
4. **Attract organic particles.** Hold the charged dish ~5 cm above the spread soil and move it slowly. Organic particles will cling audibly and visibly. Tap the dish gently with forceps to release any attached mineral particles.
5. **Rinse into cup.** Rinse collected particles from the petri dish into a pre-labeled 100 mL cup using a DI water squirt bottle.
6. **Repeat.** Remix and redistribute the soil. Repeat steps 3–5 with a fresh petri dish until no particles are attracted.
7. **Recombine fractions.** Add the >250 µm fraction back to the <250 µm fraction in the tin. Mix well. Proceed to Part II.

> ⚠ **NOTE:** If humidity is high, dry the soil briefly with the hair dryer before attracting. Excessive humidity prevents attraction.
> *Large visible organic particles can also be removed with forceps.*

PART II — Wet Free POM & Dissolved OM (DOM)

Goal: From 10 g of the cleaned bulk soil, isolate (1) Wet Free POM >53 µm by pipette-wet-sieving, and (2) Dissolved OM (DOM) <0.45 µm by shaking + centrifugation + filtration. The residue pellet proceeds to Part III.

Materials

- 250 mL centrifuge tube (1/sample)
- 400 mL beaker (1/sample)
- 500 mL filter flask (1/sample)
- 500 mL Nalgene bottle (1/sample)
- Scintillation vial (1/sample)
- 0.45 µm membrane filter (1/sample)
- 53 µm sieve (3" diameter)
- Analytical balance
- Vortex mixer
- Centrifuge
- Orbital shaker
- Filtration setup (vacuum or gravity)
- 10 mL pipette + tips
- Small brush or spatula

A — Separate Wet Free POM by Pipette-Wet-Sieving

8. **Weigh soil.** Transfer 10.00 g of recombined bulk soil to a labeled 250 mL centrifuge tube. Record tube tare weight.

9. **Add water.** Pour 50 mL DI water slowly down the inner wall of the centrifuge tube to minimize aggregate disruption. (The remaining 50 mL will be added in step A13.)
10. **Settle.** Shake gently by hand 5 times. Let settle for 5 minutes.
11. **Pipette supernatant.** Set a 53 µm sieve over a 400 mL beaker. Using a 10 mL pipette, aspirate the supernatant and any floating organic particles from the top down to the surface of the sedimented soil. Dispense over the 53 µm sieve.
12. **Collect dense particles.** Add a few mL of DI water from the remaining 50 mL. Tilt the tube slowly so water passes over the sedimented surface. Pipette up the dislodged particles and dispense over the sieve.
13. **Rinse POM onto sieve.** Using the remaining 50 mL of DI water, wash all material retained on the sieve into a pre-labeled 100 mL cup (Wet FPOM).

⚠ **NOTE:** A small brush can help dislodge organic material clinging to the tube wall. Rinse bristles carefully — do not contaminate the sample with brush fibers.
Wet FPOM can later be sub-sieved into coarse (>250 µm) and fine (53–250 µm) fractions if needed.

B — Extract Dissolved OM (DOM)

14. **Target weight.** Target mass = tube tare + 10 g soil + 100 g water. Add the sieved supernatant from the beaker back to the centrifuge tube, then top up with DI water to reach the target weight.
15. **Shake.** Shake on an orbital shaker for 12 hours.
16. **Centrifuge.** Centrifuge for 30 minutes at 10,000 × g.
17. **Filter DOM.** Decant supernatant into a 500 mL filter flask. Filter through a 0.45 µm membrane filter into a labeled 500 mL Nalgene bottle. Discard material retained on filter.
18. **Record volume.** Measure and record the filtrate volume.
19. **Store.** Refrigerate DOM ≤2 weeks, or freeze-dry for longer storage.
20. **Retain pellet.** Keep the soil-extraction residue in the centrifuge tube for Part III.

⚠ **NOTE:** Do not store filtered DOM in the refrigerator for more than 2 weeks. Freeze-dry subsamples if longer storage is needed.
Centrifuge g-force remains constant across all centrifugation steps in this protocol.

PART III — Occluded POM (OPOM) & MAOM

Goal: Apply 60 J/cm³ ultrasonication to the Part II extraction residue to disrupt macro-aggregates, then wet-sieve to separate (1) Occluded POM (OPOM) >20 µm and (2) Mineral-Associated OM (MAOM) <20 µm.

Materials

- 50 mL beaker (1/sample)
- 400 mL beaker (1/sample)
- 1000 mL beaker (1, for ice bath)
- 100 mL cup (1/sample, labeled OPOM)
- 20 µm sieve (3" diameter)
- Ice
- Analytical balance
- Vortex mixer
- Sonicator with 1 cm probe

Steps

21. **Transfer residue.** Add 5 mL DI water to the centrifuge tube. Vortex ~10 seconds to re-suspend the pellet. Pour into a labeled 50 mL beaker. Rinse the tube with minimal additional DI water.
22. **Top up to 30 mL.** Using the target-weight approach (beaker tare + 10 g soil + 30 g water), add DI water to reach 30 mL total.
23. **Ice bath.** Place the 50 mL beaker in an ice bath inside a 1000 mL beaker.

24. **Sonicate.** Insert probe 1 cm. Sonicate at 60 J/cm³ (39 seconds). See note for time calculation.
25. **Rinse probe.** Rinse soil off the sonication probe back into the beaker with DI water. Record final volume.
26. **Settle 5 minutes.**
27. **Sieve.** Set a 20 µm sieve over a labeled 400 mL beaker (MAOM). Pour the suspension through the sieve. Rinse particles retained on the sieve with DI water into the MAOM beaker from underneath the sieve.
28. **Collect OPOM.** Concentrate material remaining on top of the sieve to one side using DI water. Transfer into a pre-labeled 100 mL cup (OPOM).
29. **Freeze.** Store OPOM and MAOM cups in the freezer for freeze-drying.

> ⚠ **NOTE:** Sonication time formula: t (s) = [Target energy (J/cm³) × volume (mL)] ÷ [power (W)]. Recalculate if volume changes.
>
> *The ice bath is essential to prevent sample heating during sonication.*

PART IV — Sample Analysis

Solid Fractions (Dry FPOM, Wet FPOM, OPOM, MAOM)

Freeze-dry all solid fractions. Weigh and transfer to labeled whirl-pak bags. Fractions can be analyzed for:

- Total C, Total N, ^{15}N
- Organic C, ^{13}C, ^{12}C
- Radiocarbon (^{14}C)

Liquid Fraction (DOM)

Thaw frozen DOM fractions before analysis. DOM fractions can be analyzed individually or combined. If combining, use equal volumes (e.g., 100 mL each) to ensure equal representation. Fractions can be analyzed for:

- Dissolved organic carbon (DOC)
- Exchangeable Ca^{2+}